M.Sc. Chemistry Curriculum Structure

Introduction to the Program

Chemistry involves the study of substances, their properties, structures, reactions and transformations. Chemistry is interlinked with various scientific disciplines and play an important role in our daily life. It broadly covers various areas including medicinal chemistry, analytical chemistry, organometallic chemistry, polymer chemistry, quantum chemistry, catalysis and many aspects of materials science. In the modern society, knowledge of Chemistry is essential for addressing most of the real-life challenges related to water, energy, food, agriculture, security and healthcare.

Objectives of the Program

The objectives of the program are to bring out professionals having fundamental knowledge; specific skill-sets of various aspects of chemistry such as Organic Synthesis, Main-Group and Transition Metals, Organometallics and Catalysis, Physical and Spectroscopic techniques, Inorganic and Solid-State Material synthesis, Computational and Theoretical Chemistry.

The department focuses on preparing future students to take up challenges at various levels of academics, R&Ds and industries.

Expected Graduate Attributes

- Capability to appreciate synthetic methods for development of important chemical products such as pharmaceutical drugs, paints, petrochemicals, food additives, plastics etc.
- Ability to use characterization tools for analysing reactions and monitoring chemical processes for product development.
- Comprehend analytical, spectroscopic, microscopic, and other instrumental tools for utilizing in interdisciplinary research.
- Use of computational tools to understand and analyse chemical and biophysical processes, molecular structures and advanced materials.
- Develop written and oral communication skills to write and present scientific articles and technical reports.
- Ability to plan and manage technical research projects and work in a team.
- Appreciation and adherence to norms of professional ethics and sense of intellectual integrity.

Program Learning outcome

- Strong fundamentals knowledge in both theoretical and practical aspects of organic, inorganic, and physical chemistry.
- Acquire experimental data handling skill-sets to address cutting-edge technological challenges across diverse domains.
- Develop the ability to exercise independent learning, thinking and problem-solving skills.
- Ability to interact in a cross-cultural environment and acquire time and project management skills for carrying out guided research work in any of the research area related to science and technology.
- Cultivate strong communication skills to present scientific learning and ideas effectively.

Program Compulsory (MC)	14	42 **Credits**
Program Elective (PE)	5	**15 Credits**
Open Elective (OE)	1	**3 Credits**
Total of Non-Graded (Industry-Academia summer Internship), Academic internship allowed in IIT Jodhpur		**2 Credits**
Thesis		**12 Credit**
	Total	**74 Credits**

Course Curriculum Structure

Table 1. Semester-wise plan for MSc in Chemistry

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	I Semester						**II Semester**				
MC	Organic Reactions and Mechanisms	3-0-0	3	-	3	MC	Organic Synthesis	3-0-0	3	-	3
MC	Physical Organic Chemistry	3-0-0	3	-	3	MC	Industrial Catalysis	3-0-0	3	-	3
MC	Main Group and Bioinorganic Chemistry	3-0-0	3	-	3	MC	Solid State and Materials Chemistry	3-0-0	3	-	3
MC	Coordination and Organometallic Chemistry	3-0-0	3	-	3	MC	Heterocyclic Compounds and Applications	3-0-0	3	-	3
MC	Numerical Methods and Group Theory	3-0-0	3	-	3	MC	Statistical Thermodynamics	3-0-0	3	-	3
MC	Quantum Chemistry and Applications	3-0-0	3	-	3	MC	Principles of Spectroscopy	3-0-0	3	-	3
MC	Organic and Inorganic Chemistry lab	0-0-6	6	-	3	MC	Physical and Computational Chemistry Lab	0-0-6	6	-	3
	Total				**21**		**Total**				**21**

	Summer Semester										
	Industry-Academia summer Internship/ Academic internship									2	
	Total									**2**	
	III Semester						**IV Semester**				
ME	Program Elective-1	3-0-0	3	-	3	ME	Program Elective-4	3-0-0	3	-	3
ME	Program Elective-2	3-0-0	3	-	3	ME	Program Elective-5	3-0-0	3	-	3
ME	Program Elective-3	3-0-0	3	-	3	MO	Open Elective-1	3-0-0	3	-	3
T	Thesis	0-0-6	-	-	6	T	Thesis	0-0-6	-	-	6
Total					**15**	**Total**				**-**	**15**

Total Credits: 74

	M.Sc. in Chemistry (Elective Courses)				
1	PE	Frontiers in Organic Chemistry	3-0-0	3	700
2	PE	Asymmetric Catalysis and Applications	3-0-0	3	700
3.	PE	Supramolecular Chemistry and Applications	3-0-0	3	600
4.	PE	Chemistry of Organic Biomolecules	3-0-0	3	600
5	PE	Spectroscopic Determination of Organic Compounds	3-0-0	3	700
6	PE	Frontiers in Organometallic Catalysis	3-0-0	3	700
7	PE	Catalysis for Energy	3-0-0	3	600
8	PE	Characterization Techniques in Inorganic Chemistry	3-0-0	3	700
9	PE	Crystallography and X-ray Techniques for Chemists	3-0-0	3	700
10	PE	Inorganic Rings, Clusters and Polymers	3-0-0	3	600
11	PE	Advance Fluorescence Microscopy Technique and Application	3-0-0	3	700
12	PE	Chemical Binding	3-0-0	3	600
13	PE	Molecular Dynamics Simulations	3-0-0	3	700
14	PE	Modern NMR Methods for Chemists	3-0-0	3	600
15	PE	Principles of Biological Soft Matter	3-0-0	3	600

Title	**Organic Reactions and Mechanisms**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Pre-requisite			

Objective

The objective of the course is:

1. To familiarize students with organic reactions and provide deeper insight into the reaction mechanisms.
2. To broaden their understanding of organic chemistry with emphasis on the efficiency and selectivity of organic reactions.

Learning Outcomes

The students are expected to:

1. Be able to propose mechanisms for complex organic reactions.
2. Understand and predict the orbital interactions involved in organic reactions.

Contents

Introduction: Organic Reactions and their mechanisms; methods for determining reaction mechanisms; transition states and reactive intermediates; crossover experiments; molecular orbital interactions in common functional groups **(7 lectures)**.

Acid and base catalysis: The pK_a scale and its interpretation; pK_a of acids; pK_b (pK_aH) of bases; general and specific acid-base catalysis **(3 lectures)**.

Classification of reactions: Substitution, elimination and addition reactions; aromaticity; electrophilic and nucleophilic aromatic substitution reactions **(9 lectures)**.

*Selectivity in organic reactions: C*hemoselectivity, regioselectivity, enantioselectivity, diastereoselectivity, stereoselectivity, and stereospecificity in organic reactions; chiral pool synthesis **(8 lectures)**.

Radical Chemistry: Free radicals, radical anions, radical cations; radical initiators; radical halogenation of alkanes; Named reactions involving radical intermediates **(5 lectures)**.

Rearrangements and migrations: Cationic and carbanionic rearrangements; carbene and carbenoid-based rearrangements **(7 lectures)**.

Text Books:

1. Sykes, P., (2013), A Guidebook to Mechanism in Organic Chemistry, 6th Edition, Longman
2. Clayden, J., Greeves, N., and Warren, S., (2012), Organic Chemistry, 2nd Edition, Oxford
3. Carey, F.A., and Sundberg, R.J. (2007). Advanced Organic Chemistry, Part A: Structure and Mechanisms, 4th Edition, Plenum

References:

1. Gossman, R.B. (2008), The Art of Writing Reasonable Organic Reaction Mechanism, Springer

Self-learning Material:

1. Sunoj, R.B., Organic Reaction Mechanism, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay, http://nptel.ac.in/courses/104101005/

Title	**Physical Organic Chemistry**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Pre-requisite			

Objective

The instructor will:

1 To provide physical concepts of organic chemistry
1. To provide knowledge in pericyclic and photochemical reactions

Learning Outcomes

The students are expected to:

1. Be able to understand the course of organic reactions with regard to physical concepts and stereochemical models
2. Develop an in-depth knowledge with mechanistic understanding in pericyclic and photochemical reactions and apply those in the synthesis of organic compounds

Contents

Introduction: Brief review of basic physical organic concepts **(2 lectures)**.

Stereoelectronic Effects: Participation of sigma and pi bonds in transition state, Orbital symmetry and frontier orbitals, Hammond's Postulate, Hammette Plot, Curtin–Hammett principle **(6 lectures)**.

Stereochemical Models: Cram Model, Felkin-Anh Model, Chelation Model, Prelog's Model, Zimmerman-Traxler Model **(7 lectures)**.

Pericyclic Reactions: Introduction of pericyclic reactions, Classifications, Cycloaddition, Electrocyclic, Sigmatropic, Group transfer reactions **(14 lectures)**.

Photochemistry: Basic principles of photochemistry, Retinal isomerization, Photochemical reaction of cyclic and acyclic carbonyl compounds, including enones and dienones, photochemistry of alkenes, dienes, aromatic compounds **(10 lectures)**.

Text Books:

1. Anslyn, E. V. and Dougherty, D. A., (2006), Modern Physical Organic Chemistry, University Science Books
2. Singh, J. and Singh, J. (2012). Photochemistry and Pericyclic Reactions, 3rd Edition. New Age

References:

1. Smith, M. B., (2016), March's Advanced Organic Chemistry, 7th Edition. Wiley
2. Carey, F. A. and Sundberg, R. J., (2007), Advanced Organic Chemistry Part A: Structure and Mechanisms, 5th Edition. Springer
3. Carey, F. A. and Sundberg, R. J. (2007), Advanced Organic Chemistry Part B: Reactions and Synthesis, 5th Edition. Springer

Self-learning Material:

1. Singh, N.D. P., Organic Chemistry and Pericyclic Reactions, NPTEL Course material, Department of Chemistry, Indian Institute of Technology Kharagpur, http://www.nptel.ac.in/courses/ 104105071/

Title	**Main Group and Bioinorganic Chemistry**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Give better insight into chemistry of main group elements and bioinorganic chemistry
2. Provide broad knowledge about properties and applications of compounds of main group elements and metal ions essential in biological systems.

Learning Outcomes

The students are expected to have the ability to:

1. Appreciate chemistry of elements essential in biological systems.
2. Understand structure and bonding concepts in chemistry of s- and p-block elements and systematic understanding of their chemical reactivity

Contents

Chemistry of main group elements: Molecular Geometry and Symmetry, hypervalency, Synthesis, structure and bonding in polyhedral boranes and carboranes, Wade's Rules, Berry Pseudorotation, electron count, Isolobal analogy. B-N, P-N, S-N unit containing compounds. Compounds of Silicon, Zeolites. Hydrides, halides, oxides, nitrides of s- and p- block elements. Allotropes of Carbon, introduction to carbenes and their heavier analogues. Chemistry of halogens and Noble gases, Chemistry of s-block elements. Organometallic compounds of main group elements. (**15 Lectures**)

Spectroscopic Characterization: NMR (Including temperature (VT) NMR, e.g., PF_3Cl_2), FT-IR, Raman, Mass etc. (**4 Lectures**)

Applications of main group compounds: Frustrated Lewis pairs and small molecule activation, Synthetic uses of main group complexes, Oxidizing/reducing agents. Precursors for materials deposition. Maingroup based polymers. Deposition of thin films of materials, with a strong focus on Chemical Vapour Deposition (CVD) and ALD. PET and SPECT imaging. (**8 Lectures**)

Bioinorganic Chemistry: Transition metals in biology - their occurrence and function, active-site structure, Metalloenzymes; O2 binding properties of heme (hemoglobin and myoglobin) and non-heme proteins hemocyanin & hemerythrin), co-operativity effect, Bohr Effect; representative synthetic models of heme and nonheme systems. Electron transfer proteins - active site structure and functions of ferredoxin, rubredoxin and cytochromes, and their comparisons. Metals in medicine. (**12 Lectures**)

Textbook

1. 1. Huheey, J. E., (2008), *Inorganic Chemistry*, 4th edition, Pearson
2. Housecroft, C. E., and Sharpe, A. G., (2012), *Inorganic Chemistry*, 4th Edition, Pearson
3. Miessler G. L., (2014), *Inorganic Chemistry*, 4th edition, Pearson

Reference Books

1. S. J. Lippard (1994) Principles of Bioinorganic Chemistry, University Science Books.
2. Norman, N. C., (2014), *Periodicity and the p-Block Elements*, Oxford Primer Nos. 51,2nd Edition, Oxford University Press
3. Atkins, P. (2010) *Inorganic Chemistry*, 5th Edition, Oxford University Press

Self Learning Material

1. Balakrishna, M. S., *Chemistry of Main Group Elements*, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay, http://nptel.ac.in/courses/117101105/
2. Ray, D., *Bioinorganic Chemistry*, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology, Kharagpur, http://nptel.ac.in/courses/104105031/

Title	**Coordination and Organometallic Chemistry**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Provide background to basic concepts and key knowledge base of Coordination and Organometallic Chemistry.

Learning Outcomes

The students are expected to have the ability to:

1. Understand the bonding fundamentals for both coordination compounds, predicting geometries of simple molecules.
2. Appreciate the fundamentals of organometallic chemistry.

Contents

Introduction: Coordination geometry, coordination numbers, ligands, isomerism, thermodynamic stability, step-wise and overall binding constant, chelate and macrocyclic effect. (**3 Lectures**)

Theories of Bonding: VBT, CFT, MOT and their limitations; splitting of d-orbitals. Spectral and magnetic properties; Color of complexes, term symbols, selection rules for electronic transitions, Orgel and TS diagram, and Nephelauxetic effect, magnetochemistry, Jahn-Teller theorem. (**11 Lectures**)

Inorganic reactions and mechanisms: Hydrolysis reactions, substitution reactions trans-effect; isomerization reactions; metal-metal bonding, cluster compounds (**5 Lectures**)

Transition metal organometallics: Introduction: various sigma/pi donor acceptor ligands, 18 electron rule isolobal analogie. Metal carbonyls, metal phosphines, olefin and acetylene complexes, alkyls and allyl complexes, metallocenes, carbenes and NHC complexes Major reaction types – oxidative addition, reductive elimination, migratory insertion, elimination isomerization and rearrangement reactions. **(12 Lectures)**

Chemistry of f-block elements: Introduction, Structure and bonding, Shift reagents, strong magnets, Luminescence, Separation and Extraction Techniques. Organometallic compounds of f-block elements: synthesis structure bonding and properties **(5 Lectures)**

Organometallic compounds in medicinal and material chemistry **(3 Lectures)**

Textbook

1. Huheey, J.E. Keiter, E. A.; Keiter, R. L.; Medhi, O.K.; (2006), Inorganic Chemistry, Pearson; 4th edition
2. Cotton, F. A.; Wilkinson, G.; Basic Inorganic Chemistry (2007), Wiley; 3rd edition

Reference Books

1. Housecroft, C and Sharpe, A. G. (2012) Inorganic Chemistry, 4th Edition, Pearson

Self-Learning Material

1. Maiti, D., Basics in Inorganic Chemistry, NPTEL Course Material,Department of Chemistry, Indian Institute of Technology, Bombay, https://www.youtube.com/watch?v=0ofu2inFF0k&list

Title	**Numerical Methods and Group Theory**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Prerequisite			

Objectives

1. Basic mathematics and numerical techniques required for Chemistry students
2. Knowledge on error analysis to be used for laboratory courses
3. Provide the fundamentals of group theory to understand structure, chemical bonding and spectroscopy

Learning Outcomes

1. Ability to analyze data and perform error analysis on the data
2. Ability to understand and appreciate the mathematical and group theoretical concepts behind chemical theories

Course Content

Mathematical Concepts: Vectors and Vector spaces, Operators, Orthogonal and Unitary matrices, Diagonalization, Matrix eigenvalue problems, Systems of linear algebraic equations, Determinants, Ordinary and partial differential equations, Fourier and Laplace transformation **(13 Lectures)**

Numerical Methods: Errors in Data, Absolute and relative errors, Distribution of Errors, Central Limit Theorem, Linear regression and correlation, Interpolation, Roots of Equations, Algorithms for Matrix **(13 Lectures)**

Group Theory: Symmetry elements and Point Groups, Group postulates and definitions, Group multiplication tables, Subgroups and cosets, Symmetry classes and conjugates, Reducible and irreducible representations, Representations and character tables, Group theory and quantum mechanics, Group theory and chemical bonding, Applications of group theory to structures and spectroscopy, Group theory and crystal symmetry **(13 Lectures)**

Text Books

1. Kreyszig, E., (2011) *Advanced Engineering Mathematics*, 9th Ed., Wiley
2. Cotton, F. A., (2008) *Chemical Applications of Group Theory*, 3rd Ed., Wiley India

Reference Books:

1. Louis, L., (1991) *A Practical Guide to Data Analysis for Physical Science students*, Cambridge University Press
2. Louis, L., (2005) *Mathematics for Science Students*, Cambridge University Press
3. Arfken, W. and Harris, P. (2012) *Mathematical Methods for Physicists*, 7th Ed., Elsevier
4. Bishop, D. M., (1993), *Group Theory and Chemistry*, 2nd Ed., Dover Publications, New York

Online Course Material:

1. Iyengar, S. R. K., *Numerical Methods and Computation,* NPTEL course material, Department of Mathematics, IIT Delhi
2. https://www.youtube.com/watch?v=88ys5ZIolSg&list=PL6E313980EF23CA6E
3. Chandra, M., *Chemical Applications of Symmetry and Group Theory,* NPTEL course material, Department of Chemistry, IIT Kanpur
4. https://www.youtube.com/watch?v=Av9f25sqLG0&list=PLj_Alq7xw30knZPTpa9whzqiSn_RZHGWP

Title	**Quantum Chemistry and Applications**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry	Type	Compulsory
Prerequisite	-		

Objectives

The Instructor will:

1. provide an advanced level understanding of quantum chemistry
2. Will show a few applications of quantum mechanics to chemistry

Learning Outcomes

The students are expected to have the ability to:

1. understand quantum mechanics and interpret chemistry problems at the atomistic level
2. apply quantum chemistry to problems in chemistry

Contents

Fundamentals: Old Quantum theory, Correspondence principle, Bohr-Sommerfeld quantization, Wave-particle duality, Stern-Gerlach experiment **(7 Lectures)**

Mathematical Formalism: Operators, Eigenfunctions and eigenvalues, Operators in quantum mechanics, expectation values, Theorems of Quantum Mechanics, Commuting and non-commuting operators, Angular momentum **(8 Lectures)**

Exactly Solvable Problems: Time independent and time dependent wave equation, Particle confined to infinite and finite potential wells, Harmonic oscillator, Rigid rotor, Hydrogen atom **(8 Lectures)**

Approximation Methods: Variational principle, Perturbation theory, Energy and wavefunction corrections **(8 Lectures)**

Quantum Chemistry: Spin, Many-electron problems, Hartree-Fock method, Born-Oppenheimer approximation **(8 Lectures)**

Textbook

1. Quantum Chemistry, I. N. Levine, 7th Edition, PEARSON (2016).

Reference Books

1. Quantum Chemistry, D. A. McQuarrie, Viva Student edition (2016).
1. 2. Modern Quantum Chemistry, A. Szabo and N. S. Ostlund, Dover Books (1996).

Title	**Organic and Inorganic Chemistry Lab**	Course No.	CYP4XX0
Department	Chemistry	L-T-P [C]	0–0–6 [3]
Offered for	M.Sc. Chemistry	Type	Compulsory
Prerequisite			

Objectives

The instructor will:

1. Give students a practical understanding of the theoretical phenomenons.
2. Provide an exposure to the laboratory techniques for the purification and isolation of organic compounds
3. Emphasize on the different techniques of reaction set-up (air-sensitive, moisture-sensitive etc. involved in the synthesis, isolation and purification of inorganic compounds).
4. Expose students to various spectroscopic identification and characterization techniques.

Learning Outcomes

The students are expected to:

1. Learn basic experimental techniques in organic chemistry
2. Be able to carry out simple synthetic organic transformations and isolate the reaction products
3. Understand the basic principles behind the synthesis and properties of inorganic compounds.
4. Plan and Conduct experiments for synthesis and characterizing inorganic compounds.

Contents

1. Extraction: separation of an acidic, a basic and a neutral substance
2. Thin Layer Chromatography (TLC): extraction and TLC of spinach pigments
3. Hands on Reactions and Purifications by Column Chromatography: Wittig olefination, Henry reaction (Nitro-Aldol reaction), Grignard reaction etc..
4. Distillation: Separation of compounds based on boiling point
5. Elimination Reaction: Dehydration of methylcyclohexanols to methylcyclohexene
6. Cycloaddition: The Diels-Alder reaction of a conjugated diene either in eucalyptus oil or other dienophiles
7. Aromatic Substitution: The Friedel-Crafts acylation of ferrocene
8. Functional Group Interconversion: Reduction and oxidation of organic compounds
9. Drug Synthesis: Synthesis of Aspirin from Salicylic Acid
10. Green Chemistry: A Solvent-Free Aldol Condensation
11. Non-metal complex: Synthesis and characterization of bispyridine iodide nitrate;
12. Solid phase synthesis and characterization of trans-bis glycinato copper(II);
13. Synthesis of penta amminechlorocobalt(III)chloride and its spectroscopic properties;
14. Influence of ligand field tetragonality on the ground state spin of nickel(II) complexes with NH3 and H2O and ethylene diamine;
15. Preparation and magnetic properties of FeIII(acac) complex;
16. Preparation of TiO2 from Titanium Isopropoxide;
17. Preparation and Characterization of Hydroxyapatite;
18. Luminescent Ruthenium Complexes. Synthesis of RuHCl(CO)(PPh3)3 complex and its characterisation by 1H, 31P NMR and IR spectroscopy
19. FLP H2 activation and hydrogenation reactions
20. Synthesis of MOF (HKUST or ZIF-8).

Textbook

1. Vogel, A. I., *Textbook of Practical Organic Chemistry*, Pearson

2. G. S. Girolami, T. B. Rauchfuss, R. J. Angelici Synthesis and Technique in Inorganic Chemistry, University Science Books, 3rd Ed, 1999.
3. Elias, A. J., A Collection of Interesting General Chemistry Experiments, Universities Press(India) Pvt. Ltd., 2002

Reference Books

1. Ault, A., *Techniques and Experiments for Organic Chemistry*, University Science Books
2. J. Derek Woolins, Inorganic Experiments, 3rd Edition, Wiley, 2010.
3. Roesky, H. W.; Muckel, K., Chemical Curiosities: spectacular experiments and inspired quotes, VCH, 1996.
4. Handouts prepared for the laboratory experiments: collections from various literature sources (ACS Chemical Education)

Self Learning Material

John Dolhun. 5.301 Chemistry Laboratory Techniques. January IAP 2012. Massachusetts Institute of Technology: MIT Open Courseware, https://ocw.mit.edu.

Title	**Organic Synthesis**	Course No.	CYL4XXX
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Pre-requisite			

Objective

The instructor will:

1. To provide an understanding of diverse chemical reactions and their application in synthesis
2. To provide knowledge of key aspects of retrosynthesis of simple-to-complex organic molecules.

Learning Outcomes

The students are expected to:

1. Acquire the knowledge and skills required to strategize and plan organic synthesis,
2. Develop an understanding of total synthesis of organic molecules.

Contents

Functional group interconversions and Name reactions: Principles of functional group interconversions; Reactivity patterns of common functional groups; Carbon–carbon and carbon–heteroatom bond-forming reactions; Important name reactions in organic synthesis; Mechanistic aspects and synthetic applications of selected name reactions **(10 lectures)**.

Transition Metal-Catalyzed Reactions in Organic Synthesis: Introduction to organometallic and transition metal chemistry; Transition metal-catalyzed cross-coupling reactions; Palladium-, nickel-, and copper-catalyzed transformations; Fundamental organometallic processes (Oxidative addition, Reductive elimination, Transmetallation, Migratory insertion) Catalytic cycles and mechanistic pathways; Synthetic applications of cross-coupling reactions in modern organic synthesis **(9 lectures)**.

Oxidation and Reduction Reactions in Organic Synthesis: Fundamentals of oxidation and reduction reactions; Oxidation states and redox concepts in organic chemistry; Common

oxidizing reagents and their synthetic applications; Common reducing reagents and selective reductions; Chemoselective and stereoselective redox transformations; Name reactions involving oxidation and reduction processes; Applications of redox reactions in multistep organic synthesis **(8 lectures)**.

Carbanions and Organometallic Reagents in Synthesis: Generation, stability, and reactivity of carbanions; Alkylation reactions of enolates; Synthetic applications of enamines and hydrazones; Alkylation of heteroatom-stabilized anions; Organometallic reagents: preparation and properties; Applications of organolithium, Grignard, and related reagents in organic synthesis; Carbon–carbon bond formation using organometallic intermediates **(7 lectures)**.

Strategy and Design in Organic Synthesis: Principles of retrosynthetic analysis; Synthetic planning and disconnection approaches; Concepts of synthetic equivalents; Umpolung and polarity inversion strategies; Design and execution of efficient multistep organic syntheses **(5 lectures)**.

Text Books:

1. Warren,S., and Wyatt.P., (2008), Organic Synthesis: The Disconnection Approach, 2nd Edition, Wiley-VCH
2. Carey,F.A., Sundberg,R.J., (2007), Advanced Organic Chemistry, Part B: Reactions and Synthesis, 5th Edition, Plenum
3. Carruthers, W., (1989), Some Modern Methods in Organic Synthesis, Cambridge
4. Clayden,J., Greeves,N., and Warren,S., (2012) Organic Chemistry, 2nd Ed., Oxford

References:

1. Nicolaou,K.C., Sorensen,E.J., (1996), Classics in Total Synthesis: Targets, Strategies and Methods, 1st Edition, Wiley-VCH

Self-Learning Material

1. Punniyamurthy,T., Principles of Organic Synthesis, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Guwahati, http://www.nptel.ac.in/courses/104103022/

Title	**Heterocyclic Compounds and Applications**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Pre-requisite			

Objective

The instructor will:

1. Provide insight into the fundamental reactivity and synthesis of heterocycles, and
2. Broaden their knowledge in the field of heterocyclic chemistry with regard to its application in the synthesis of natural products and pharmaceuticals.

Learning Outcomes

Students will be able to:

1. Understand the basic heterocyclic principles and apply different types of organic reactions in the synthesis of small heterocyclic compounds.
2. Learn the applications of modern organic synthetic tools in the efficient synthesis of heterocycles.

3. Design and propose routes for the synthesis of natural products and pharmaceuticals based on heterocycles.

Contents

Introduction: Importance of heterocyclic compounds, Nomenclature of heterocyclic ring systems **(3 lectures)**.

Three-membered Heterocycles: Structure, reactivity and synthesis of Oxirane, Thiirane, and Aziridine **(3 lectures)**.

Four-membered Heterocycles: Structure, reactivity and synthesis of four membered Heterocycles such as Oxetane, Thietane and Azetine **(3 lectures)**.

Five-membered Heterocycles: Structure, reactivity and synthesis of five membered Heterocycles such as Pyrrole and pyrrolidine, Thiophene and tetrahydrothiophene, Furan and tetrahydrofuran, chemistry of Pyrazole, Imidazole, Oxazole, Thiazole **(10 lectures)**.

Six-membered Heterocycles: Structure, reactivity and synthesis of Six membered Heterocycles: Pyridine and Piperidine, pyran and Tetrahydropyran, pyridones, Pyridine-N-oxides **(10 lectures)**.

Polycyclic heterocycles: Structure, reactivity and synthesis of Indole, Quinoline and Isoquinoline, Coumarins and Chromones, benzimidazole and benzoxazole **(6 lectures)**.

Modern Methods: Synthesis of heterocycles using modern sustainable methods **(4 lectures)**.

Text Books:

1. J. A. Joule and K. Mills, Heterocyclic Chemistry, 5th Ed., Wiley, 2010.

References:

1. R. K. Bansal, Heterocyclic Chemistry, 5th Ed., New Age International Private Limited, 2017.

Self-Learning Material

1. NPTEL Lectures : Heterocyclic Chemistry- http://nptel.ac.in/courses/104105034/

Title	**Statistical Thermodynamics**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Provide the connection between microscopic theory and thermodynamics.
2. Describe how complex phenomena can be transferred into simple models.

Learning Outcomes

The students are expected to:

1. Predict properties of many-body systems starting from its microscopic constituents and their interactions
2. Connect microscopic theory to chemical reaction

Contents

Mathematical foundation: Probability theory, Permutation and combination, Conditional probability and independent events, Random variables and their probability distribution, Central limit theorem, Binomial distribution, Stirling's approximation, Method of Lagrange multiplier; Statistics of scientific measurement, Error, Maximum likelihood method, Curve fitting, Least square fit; Dirac delta function, Uniform distribution and Dirac delta function, Partial differential equation, equation of continuity (**11 Lectures**)

Statistical thermodynamics: Micro and Macro states, postulates, concepts of distinguishable and indistinguishable particles, Microcanonical Ensemble (NVE), Canonical Ensembles or Isothermal-isochoric (NVT) Ensembles, Isothermal-isobaric (NPT) and Grand Canonical Ensemble (μVT), Equivalence of Ensembles, Boltzmann distribution, Entropy, partition functions for different ensemble, calculations of thermodynamic properties of each of ensemble, ideal monatomic and diatomic gas, Applications of canonical ensemble, 1D Ising model, Ideal and real gas, Sackur Tetrode equation, Phase space, Ergodicity, Liouville's Theorem (**15 Lectures**)

Applications: Fluctuations and thermodynamic properties, Heat capacity, compressibility, Linear response theory, Partition function and Transition State Theory, classical fluids, Pair correlation functions, Response functions, Langevin Equation, Fluctuation-Dissipation Theorem, Diffusion (**13 Lectures**)

Textbook

1. Tulsi Dass and Satish K Sharma, Mathematical Methods in Classical and Quantum Physics, Universities Press 1998.
2. D. A. McQuarrie and John D Simon, Physical Chemistry, Viva Books, India 2003.
3. D. McQuarrie, Statistical Mechanics, University Science Books, 2000.

Reference Book

1. Chandler, D., Introduction to Modern Statistical Mechanics, Oxford University Press, 1987.
2. Biman Bagchi, Statistical Mechanics for Chemistry and Materials Science, CRC press, 2018.
4. Molecular Driving Forces: Statistical Thermodynamics in Chemistry and Biology by K. A. Dill and
1. S. Bromberg, Garland Science, 2003.
5. R Shankar, Basic Training in Mathematics, Springer, 2006.

Self-Learning Material

1. https://nptel.ac.in/courses/104101139
2. Course Introduction Basic Statistical Mechanics by Biman Bagchi
3. https://www.youtube.com/watch?v=M_dhvmM2fmI&list=PLOzRYVm0a65en9rkAeG_KDhh4thVEu5Od

Title	**Industrial Catalysis**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Impart knowledge on Catalysis and its importance.
2. Teach practical applications of catalysts applied in industry.

Learning Outcomes

1. The learners are expected to gain knowledge of independent thinking in designing, characterizing and fine tuning of catalysts suitable for potential catalytic transformations.
2. Learners will be able to translate fundamental catalytic knowledge to develop technology for industrial processes.
3. Learners will have an update on current trends and challenges of catalysis towards industrial application.

Contents

Introduction: Definition and concepts of catalysis, Activity, TOF, TON, selectivity, Homogeneous vs heterogeneous and biocatalysis **(4 Lectures)**

Homogeneous catalyzed Industrial processes: Hydrogenation, hydroformylation, carbonylation, wacker synthesis, oxo synthesis, C-C coupling reactions, Oligomerisation of ethylene (SHOP process), adiponitrile synthesis, polymerization, metathesis, Asymmetric catalysis, Monsanto L DOPA process, S-Naproxen, L-menthol **(14 Lectures)**

Industrial processes with Biocatalysts: Acrylamide, aspartame synthesis, pharmaceuticals, herbicides. **(4 Lectures)**

Heterogeneously catalyzed processes: Fischer Tropsch synthesis, water gas shift reaction, methanol synthesis, oxidation of propene. Refinery processes and petrochemistry. **(11 Lectures)**

Electrochemical reactions: Methanol oxidation, Catalytic processes with renewable materials, biofuels, biorefinery, phase transfer catalysis, catalysis reactor **(6 Lectures)**

Textbook

1. Hagen, J. Industrial Catalysis: A Practical Approach

Reference Books

1. Bhaduri S.; Mukesh, D. Homogeneous Catalysis: Mechanisms and Industrial Applications

Self-Learning Material

1. Industrial homogeneous Catalysis: Principles and Mechanisms by prof. debabrataMaiti, IIT Bombay, https://www.youtube.com/watch?v=GzXpURcjJwE&list=PL_uPpV3UHww2wyLk3XpR7LBqbR_2k1Fi0

Title	**Solid State and Materials Chemistry**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Provide an overview of the relationships between molecular or solid state structures and material properties.
2. Provide an interdisciplinary understanding of solid state chemistry which aims to provide an understanding of how molecular structure affects the properties of materials.

Learning Outcomes

The students are expected to have the ability to:

1. Obtain required knowledge for understanding material science problems and structure of solids.
2. Develop insight into electronic structure of crystals and compare it with the electronic structure of nanomaterials, chemical-physical fundamentals as well as basic method of characterisation of solids.

Contents

Structure of Solids: Crystalline and Amorphous, diffraction techniques, symmetry and point groups, packing in solids, classification, lattice energy, bonding, structures: NaCl, TiO_2, ZnS, wurtzite, Perovskite, covalent and ionic solids. (**10 Lectures**)

Crystal defects: non-stoichiometry, cluster, diffusion, Fick's Law and Kirkendall effect, Identification of defects using microscopic techniques. (**5 Lectures**)

Solid state reaction: Chemical and Physical Methods of preparation, reactivity of solids, decomposition mechanism, single crystal growth and thin film deposition. (**10 Lectures**)

Band theory: Intrinsic and extrinsic semiconductors, insulators, density of states, Dielectrics, Hall effect, Thomson, Peltier and Seeback effects. (**Lectures 8**)

Properties of Solids: Magnetic, Electrical and Optical properties, Different type of Magnetism and Superconductivity, Introduction to nanomaterials and properties. (**Lectures 6**)

Textbooks

1. B. D. Cullity and S. R. Stock, (2014), *Elements of X-ray diffraction*, 3rd edition, Pearson
2. West, A.R., (2015), *Solid State Chemistry and Its Applications*, 2nd edition, John Wiley & Sons
3. Lesley E Smart and Elaine E Moore, (2005), *Solid State Chemistry: An Introduction*, 3rd Edition, Taylor and Francis

Reference Books

1. Cheetham, A.K. and Day, P., (1997), *Solid State Chemistry Compounds*, 2nd Edition, Clarendon Press, Oxford
2. Harry R. Allcock, (2008), *Introduction to Materials Chemistry*, 1st Edition, Wiley.
3. C N R Rao and J Gopalkrishnan, (2004) *New Directions in Solid State Chemistry*, 2nd Edition, Cambridge.

Self Learning Material

Subramaniam, A, *Structure of Materials*, NPTEL Course Material, Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, http://nptel.ac.in/courses/113104014/16

Title	**Principles of Spectroscopy**	Course No.	CYL4XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor will:

1. Introduce a few spectroscopic methods used as analytical techniques in various fields of science and engineering.
2. Provide conceptual background of these methods along with their instrumentation and some applications.

Learning Outcomes

The students are expected to have the ability to:

1. Understand the conceptual background of various spectroscopic techniques
2. Connect quantum mechanics and spectroscopy to interpret experimental findings in research.

Contents

Fundamentals: Electromagnetic radiation, Maxwell's equations, Time-dependent perturbation theory, Interaction of radiation with matter, Intensity of a Transition - Transition Dipole Moment, Fermi's golden rule; Einstein treatment, Lasers and line shapes and line broadening, Laser spectroscopy; Spectroscopic Timescales, Selection rule for photon-molecule interaction, Continuous wave and Fourier Transform spectroscopy, Sensitivity and resolution of spectrum, Instrumentation, Dispersive spectrometers, Fourier Transform spectrometers (**10 Lectures**)

Rotational Spectroscopy: Linear Motion and Rotational Motion, Diatomic Rigid Rotor, Derivation of selection rules for microwave spectra, Isotope effect, Degeneracy, Intensity of Rotational Lines, Non-Rigid Rotor, Polyatomic Molecules, Rotational Raman Spectroscopy (**5 Lectures**)

Vibrational Spectroscopy: Simple harmonic oscillator, Selection rules, Anharmonicity and its effects, Rotational-vibrational spectra, Normal modes of Vibration, FT-IR Spectroscopy, Vibrational Raman spectroscopy (**7 Lectures**)

Electronic spectroscopy: Basic principle, Symmetry and Term Symbols, Born-Oppenheimer approximation, vibrational coarse and rotational fine structures, Vibrational Progression, Pre-dissociation, Selection rules from transition moment integral – Spin selection, Laporte selection, Frank-Condon principle. Jablonski diagram, various photophysical processes, quantum yield expression, Chromophores, Types of electronic transition, Effect of conjugation and solvent. Introduction to photoelectron spectroscopy. (**8 Lectures**)

Magnetic resonance Spectroscopy: Origin of spin and magnetic moment, Zeeman splitting and gyromagnetic ratio, Electron and nuclear spin resonance, Instrumentation, NMR chemical shift and spin-spin coupling, introduction to AB, AX, AMX spin system, 1D 1H and 13 NMR experiments, Pulse sequences; Methods of relaxation, Double resonance techniques; EPR – Instrumentation, First Derivative Spectra, Electron-nuclear hyperfine interaction, theory of first order EPR spectra, analysis of 1st Order spectra, example of EPR spectra of organic radicals and transition metal complexes. (**9 Lectures**)

Textbook

1. Banwell, C. N., (1994), Fundamentals of Molecular Spectroscopy, 3rd Edition, Tata McGraw-Hill
2. Book Company.
3. Bernath, P. F., (2005), Spectra of Atoms and Molecules, 2nd Edition, Oxford University Press
4. J. M. Hollas, Modern Spectroscopy, Wiley India Exclusive, 2015

Reference Books

1. Gunther, H., (1995), NMR Spectroscopy, 2nd Edition, Wiley
2. Atkin's Physical Chemistry., Peter Atkins, Julio De Paula, 7 th Edition, Oxford (2022).
3. William T. Silvast, Laser Fundamentals, Cambridge University Press.

Title	**Physical and Computational Chemistry Lab**	Course No.	CYP4XX0
Department	Chemistry	L-T-P [C]	0–0–6 [3]
Offered for	MSc in Chemistry	Type	Compulsory
Prerequisite			

Objectives

The Instructor(s) will:

1. Connect the basic principles of physical chemistry with experiments.
2. Enable students to apply theoretical understandings of various concepts to analytical techniques.
3. Provide knowledge of electronic structure calculations and molecular dynamics simulations

Learning Outcomes

The students are expected to have the ability to:

1. Design, execute and analyse experiments and their outcome.
2. Operate various analytical instruments available in the Institute.
3. Perform electronic structure calculations and molecular dynamics simulations

Contents

Spectroscopy:

1. UV-Vis spectroscopy to determine concentration of protein - application of additive nature of absorbance
2. Study protein chemical denaturation using UV-Vis spectroscopy - Calculation of extent of protein unfolding
3. Estimation of silica surface coverage by bromobenzene using FTIR
4. Quantification of ligand-protein binding interaction - determination of binding constant and bound population of ligand using Fluorescence quenching analysis

Conductivity and pHmetry:

5. Determination of CMC of SDS/CTAB and quantification of Gibbs Free energy of micellization using Conductometry
6. Determine the dissociation constant (Ka) of a weak base (NH_4OH 0.1 N) by conductometric titration using a standard solution of HCl (0.1 N).
7. Determination of pI of amino acid using pH-metry.
8. Determination of strength of HCl using pH-metric titration by NaOH

Potentiometry:

9. To determine the dissociation behavior and concentration of a given polybasic acid by potentiometric titration with standard sodium hydroxide (NaOH), and to locate successive equivalence points.
10. To determine the concentration of ferrous ions (Fe^{2+}) in a given solution by potentiometric titration against standard potassium permanganate ($KMnO_4$).

Electrochemistry

11. Determination of EMF for K2Cr2O7
12. Study of Redox Reactions and Reversibility Using Cyclic Voltammetry

Simple titration

13. Estimation of ester hydrolysis by titrating against 0.1 N NaOH
14. Determination of Free alkali in detergent by titrating against 0.1 N HCl
15. Estimation of carbonate and bicarbonate in a mixture by titrating against 0.1 N HCl

Optional experiments

16. Determination of Gibbs free energy of the keto-enoltautomerism by 1H NMR
17. Determination of heat capacity and verification of Dulong-Petit's law using DSC

Computational Lab: Drawing geometry of molecules; Single-point energy calculations; Geometry optimization; Equilibrium and Saddle points; HF and DFT calculations ; Build bonded and non-bonded potential parameters after geometry optimization, Energy minimization of a bulk system at meso-scale, VMD, Run NVT, NPT.

Textbook

1. Khosla, B.D.; Garg, V.C.; Gulati, A. (2015), Senior Practical Physical Chemistry, R. Chand & Co, New Delhi
2. Kapoor, K.L. (2019),A Textbook of Physical Chemistry, Vol.7, 1st Edition, McGraw Hill Education.
3. Garland, C. W.; Nibler, J. W.; Shoemaker, D. P.(2003),Experiments in Physical Chemistry, 8th Edition, McGraw-Hill, New York.
4. Quantum Chemistry, I. N. Levine, 7th Edition, PEARSON (2016).

Reference Books

1. Ghosal, Mahapatra and Nad (2014) – An Advanced Course in Practical Chemistry, New Central Book Agency.
2. Leach, A. Molecular Modelling, 2nd Edition, PEARSON (2001).

Program Elective (PE) Courses

Title	**Frontiers in Organic Chemistry**	Number	CYL7XX0
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	M.Sc./PhD (CY) Program	Type	Elective
Pre-requisite			

Objectives

The Instructor will:

1. Give an in-depth exposure to the latest advances in organic synthesis
2. Inculcating in students a research-oriented approach towards solving synthetic organic chemistry problems

Learning Outcomes

The students are expected to:

1. Advance their knowledge in strategizing and synthesis of complex organic compounds
2. Systematically read, understand and interpret the research reported in journals
3. Develop skills of scientific presentations and interpersonal skills

Contents

Protecting Group: Selectivity and requirement of protecting groups, protecting groups and their deprotection **(8 lectures)**

Advanced Radical Chemistry: Radical substitution reactions, coupling reactions, radical mediated cyclizations, addition reaction involving radicals, radical chain reactions **(8 lectures)**

Cycloaddition Reactions: 1,3-dipolar cycloadditions, higher order cycloadditions, asymmetric cycloadditions. **(5 lectures)**

Advanced Enolate Chemistry: Diastereoselective aldol reactions, Evan's chiral auxiliary, Ender's chiral auxiliary, diastereofacial selectivity, asymmetric aldol reactions, Organocatalyzed aldol reactions **(7 lectures)**

Strategic disconnections and retrosynthetic analysis: basics of disconnection, synthons, synthetic equivalents, synthesis of small organic molecules, *Total synthesis of complex molecules, such as,* penicillin, estrone, sparteine, prostaglandins, synthesis of important pharmaceuticals **(11 lectures)**

Text books

1. Carriera, E.M., and Kvaerno,L., (2009), *Classics in Stereoselective synthesis*, Wiley-VCH
2. Kurti,L., Czako,B., (2005), *Strategic Applications of Named Reactions in Organic Synthesis*, Elsevier
3. Carey,F.A., Sundberg,R.J., (2007), *Advanced Organic Chemistry, Part B: Reactions and Synthesis*, 5th Edition, Plenum

Reference books

1. Hudlicky,T., and Reed,J.W., (2007) *The Way of Synthesis*, Wiley-VCH
2. Nicolaou,K.C., and Snyder,S.A., (2003), *Classics in Total Synthesis-II: Further Targets, Strategies, Methods*, Wiley-VCH

Self Learning Material

1. Punniyamurthy,T., *Catalytic Asymmetric Synthesis*, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Guwahati, http://nptel.ac.in/courses/104103067/

Title	**Asymmetric Catalysis and Applications**	Course No.	CYL7XX0
Department	Chemistry	Structure (L-T-P-C)	3-0-0 [3]
Offered for	MSc in Chemistry	Type	Elective
Pre-requisite			

Objective

The instructor will guide to:

1. Understand the fundamental principles of stereoselective and asymmetric synthesis/catalysis.
2. Explore various strategies and methods in asymmetric catalysis, biocatalysis, organo-catalysis, transition metal catalysis and dual catalysis.
3. Appreciate the applications of asymmetric synthesis in pharmaceuticals, agrochemicals, and industrially relevant molecules.

Learning Outcomes

Students will be able to:

1. Grasp the key principles and strategies behind asymmetric catalysis and synthesis.
2. Understand and apply different methods, biocatalysis, organocatalytic and transition metal-catalysed reactions.
3. Analyse and evaluate how asymmetric synthesis is used to create important molecules in pharmaceuticals, industry, and biology.

Contents

Principles of Selectivity: Chemoselectivity, regioselectivity, stereoselectivity, and stereospecificity. (**4 Lectures**), importance of chirality. Basics of asymmetric catalysis, including energetics of reactions.
Asymmetric Resolution: Kinetic resolution (KR), parallel kinetic resolution (PKR), dynamic kinetic resolution (DKR), dynamic kinetic asymmetric transformation (DyKAT), desymmetrization (DS). (**5 Lectures**)
Asymmetric Biocatalysis: It includes the principles of enzyme catalysis, stereoselective reactions such as enzymatic oxidation, reduction, and hydrolysis, and their applications (**5 Lectures**)
Asymmetric Metal Catalysis: The study of using transition-metal catalysts such as Rh, Ru, Pd, Ir, Ni, Co and Cu and chiral Lewis's acid catalysis. Enantioselective reduction and oxidation; Sharpless asymmetric epoxidation (**8 Lectures**)
Asymmetric Organocatalysis: Fundamentals of organocatalysts, including thiourea-based catalysts, N-heterocyclic carbenes, hydrogen-bonding catalysts, Brønsted acid and base catalysis, and phase-transfer catalysts. (**8 Lectures**)
Metal-Organo Dual Catalysis: Cooperative mechanisms, enamine, iminium, counterion, N-heterocyclic carbenes. (**5 Lectures**)
Applications of Asymmetric Catalysis: Pharmaceuticals (e.g., Thalidomide, L-DOPA), agrochemicals (e.g., Metolachlor), and industrial chiral molecules (e.g., flavors, fragrances). (**4 Lectures**)

Text Books

1. I. Ojima., (2010), *Catalytic Asymmetric* Synthesis, John Wiley & Sons, New Jersey, 3rd Ed.
2. E. N. Jacobsen, A. Pfaltz, H. Yamamoto., (1999), Comprehensive Asymmetric Catalysis III, Springer-Verlag, Berlin Heidelberg, Germany.
3. P. J. Walsh, M. C. Kozlowski., (2009), Fundamentals of Asymmetric Catalysis, University Science Books, USA.
4. R. E. Gawley, J. Aube., (2002), Principles of Asymmetric Synthesis, Springer, New York.

References Books

1. B. A. Arndtsen, L.-Z. Gong., (2020), Asymmetric Organocatalysis Combined with Metal Catalysis, Springer, Berlin.

2. G.-Q. Lin, Y.-M. Li, A. S. C. Chan., (2001), Principles and Applications of Asymmetric Synthesis, Wiley-Interscience, New York.

Self-Learning Materials

1. Prof. T. Punniyamurthy, IIT Guwahati, https://nptel.ac.in/courses/104103067

Title	**Supramolecular Chemistry and Applications**	Course No.	CYL6XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	B.S. in Engineering Chemistry and M.Sc. in Chemistry	Type	Elective
Prerequisite			

Objectives

The course aims to:

1. Develop a strong foundation in supramolecular chemistry.
2. Explain how non-covalent forces govern self-assembly and molecular recognition.

Learning Outcomes

By the end of the course, students will be able to:

1. Explain the role of non-covalent and host–guest interactions in supramolecular systems.
2. Apply supramolecular principles to advanced material design and their applications.

Contents

Fundamentals of Supramolecular Chemistry: Introduction and its historical development, non-covalent interactions, molecular recognition and complementarity, lock–key and induced fit models, positive and negative cooperativity, (**10 Lectures**)

Host–Guest Chemistry: Explores the synthesis, structure, and function of major host molecules (e.g., crown ethers, cryptands, calixarenes, cyclodextrins). Covers the governing principles of host-guest chemistry—pre-organisation, complementarity, and the lock-and-key model—in the binding of cationic, anionic, and neutral guests. (**9 Lectures**)

Self-Assembly in Supramolecular Architectures: Dynamic covalent chemistry, hierarchical self-assembly; supramolecular polymers and dynamic assemblies; molecular capsules and cages; rotaxanes and catenanes; molecular knots and mechanically interlocked molecules, stimuli-responsive systems. (**10 Lectures**)

Supramolecular Materials and Applications: Molecular machines, porous supramolecular materials; Metal–Organic Frameworks (MOFs) and covalent organic frameworks (COFs); basic principles of framework construction and topology and their applications. (**10 Lectures**)

Textbooks

1. Supramolecular Chemistry: Concepts and Perspectives, J.-M. Lehn, VCH, Weinheim, 1995.
3. Supramolecular Chemistry, J. W. Steed and J. L. Atwood, John Wiley & Sons, Chichester, 2009.

Reference Books

1. Applications of Supramolecular Chemistry, Edited By Hans-Jorg Schneider, CRC Press, 2012.

Online Resources

1. Chopra, D., Supramolecular Chemistry, NPTEL Course material, Department of Chemistry, IISER Bhopal https://nptel.ac.in/courses/104106683.
2. Bharadwaj, P. K., Supramolecular Chemistry, NPTEL course material, Department of Chemistry, Indian Institute of Technology, Kanpur, https://nptel.ac.in/courses/104104364

Title	**Chemistry of Organic Biomolecules**	Course No.	CYL6XX0
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	M.Sc. Chemistry and B.S. in Engineering Chemistry	Type	Elective
Prerequisite			

Objectives

The instructor will:

1. Introduce students to structures and functions of organic biomolecules.
2. Impart comprehensive knowledge on chemical synthesis can be applied to engineer new biomolecules for modern applications.

Learning Outcomes

The students are expected to have the ability to:

1. Describe the structures, chemical properties, and functions of biomolecules.
2. Integrate fundamental concepts to conceptualize and design chemical technology for bio-applications.

Course Contents

Introduction: Organic chemistry of living systems, pH and buffers, Covalent and non-covalent interactions in organic and biological world, Analogy between organic chemistry and biochemical reactions (**3 Lectures**)

Chemistry of Amino Acids: Structure and properties of amino acids, amide-coupling reactions, amine/acid protection and deprotection protocols, Ninhydrin test for amino acids, Synthesis of non-canonical and unnatural amino acids and applications, Peptides and solid-state peptide synthesis, Protein secondary structure, Characterization of peptides and proteins using NMR, mass, etc. (**8 lectures**)

Chemistry of Enzymes: Classification and properties of enzymes, thermodynamics and kinetics of enzyme catalysis, catalytic mechanisms, enzyme inhibition and drug design, Industrial application of a few enzymes: in organic transformation and fermentation; enzyme as medicine/drug; Abzymes in organic transformation such as Diels Alder, Baldwin's rule, Oxy-Cope rearrangement (**6 lectures**)

Chemistry of Nucleic Acids: Sugars and organic bases; Nomenclature, Conformation of sugar-phosphate backbone, Nucleotide chemistry and tautomerism, Hydrogen bonding by bases, π-stacking interactions and DNA intercalators, Chemical synthesis and application of unnatural nucleosides in modern technologies (**7 lectures**)

Chemistry of carbohydrates and fatty acids: Carbohydrates: stereochemistry, Haworth and chair notations, epimers and anomers, mutarotation, glycosidic bonds, chemical synthesis and degradation of monosaccharides, nitrogen containing sugars, applications in drug delivery systems. Structure and types of lipids, chemical synthesis of artificial lipids for modern applications. (**10 lectures**)

Chemistry of terpenes, terpenoids and steroids: Classification, chemical structures, systematic nomenclature, Isoprene rule, Biosynthesis of terpenes and cyclization mechanisms and origin of non-classical carbocation rearrangement, cholesterol and its role; Biosynthesis; application of terpenes, such as Eucalyptus oil, Isolation techniques for industrial applications (**5 lectures**)

Textbooks

1. T. W. Graham Solomons, C. B. Fryhle, S. A. Snyder. Organic Chemistry. 12th Edition, Wiley.
2. D. V. Vranken and G. A. Weiss. *Introduction to Bioorganic Chemistry and Chemical Biology*. Garland Science; 1st edition, 2012

3. H. Dugas. *Bioorganic Chemistry: A Chemical Approach to Enzyme Action*. Springer-Verlag New York Inc.; 3rd Edition, 1996

Reference Books

1. The organic chemistry of enzyme-catalyzed reactions, by Richard B. Silverman, Academic Press, San Diego, 2000, 717 pp
2. J. E. McMurry. *The Organic Chemistry of Biological Pathways*. W.H.Freeman & Co Ltd; 2nd edition, 2015

Online Learning Material

1. S. S. Bag, Bio-organic chemistry, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Guwahati, https://archive.nptel.ac.in/courses/104/103/104103018/

Title	**Spectroscopic Determination of Organic Compounds**	Course No.	CYL7XX0
Department	Chemistry	Structure (L-T-P-C)	3-0-0 [3]
Offered for	M.Sc./PhD (CY) Program	Type	Elective
Prerequisite			

Objectives

The instructor will:

1. To teach basic aspects of spectroscopic methods, important analytical tools in academic research, chemical and pharmaceutical industries for structural characterization of organic molecules.
2. Elucidation of structures of organic compounds using in-depth analysis of different spectroscopic data.

Learning Outcomes

The students are expected to:

1. Ability to determine the structures of known/unknown organic molecules utilizing several spectroscopic techniques mentioned in the content.

Contents

Infrared Spectroscopy: Introduction to IR spectroscopy, Modes of stretching and bending, stretching frequencies of different organic functional groups, Effects of structural and conformational environment (3 lectures)

Mass Spectrometry: Principles of mass spectrometry, Different ionization techniques (EI, CI, ESI etc.), GC-MS, HRMS, LC-MS and MS-MS techniques, Fragmentation and rearrangement of different classes of organic molecules, Isotope effects, Identification of known/unknown organic compounds (6 lectures)

Nuclear Magnetic Resonance Spectroscopy: Chemical shifts of different organic functional groups in ^{1}H and ^{13}C NMR, origin and interpretation of coupling constant, introduction to the principle and interpretation of DEPT, COSY, NOESY, NOE techniques, Structural elucidation of organic compounds (15 lectures)

Ultraviolet-Visible Spectroscopy: Studies on conjugated and extended conjugated systems, analysis and interpretation of UV-Visible spectroscopic data, Application in mechanistic studies (3 lectures)

Identification and Mechanistic Studies: Application of all above techniques in the structural elucidation of unknown organic compounds and investigation of organic reaction mechanisms (15 lectures)

Text Books

1. Donald L. Pavia, Gary M. Lampman, George S. Kriz, James A. Vyvyan: Introduction to Spectroscopy, 4th Edition, Brookes Cole, 2008.
2. Robert M. Silverstein, Francis X. Webster, David Kiemle: Spectrometric identification of organic compounds, 7th Edition, Wiley, 2005.

Reference Books

1. Harald Gunther: NMR spectroscopy, Basic principles, concepts, and applications in chemistry, 2nd Ed., Wiley, 2001 (reprint)
2. Timothy Claridge: High Resolution NMR Techniques in Organic Chemistry, 2nd Ed. Elsevier, 2009
3. Edmond de Hoffmann, Vincent Stroobant: Mass Spectrometry, Principles and applications, 3rd Edition, Wiley, 2007

Self Learning Material

1. http://nptel.ac.in/courses/104103071/26

Title	**Frontiers in Organometallic Catalysis**	Course No.	CYL7XX0
Department	Chemistry	L-T-P [C]	3-0-0[3]
Offered for	MSc in Chemistry	Type	Elective
Prerequisite			

Objectives

1. To impart advanced knowledge on frontiers of organometallic chemistry.
2. To teach practical applications of organometallic compounds in homogeneous catalysis particularly relevant to industrial applications.

Learning Outcomes

1. The learners are expected to gain knowledge of independent thinking in designing, characterizing and fine tuning of organometallic complexes suitable for potential catalytic transformations.
2. Learners will be able to delineate mechanistic proposal of catalytic reactions based on the available experimental data.
3. Learners will have an update on current trends and challenges of organometallic chemistry towards molecular activation and organic transformations focusing on sustainable development.

Contents

Introduction: Basics of organometallics, 18 e- rule, sigma and pi donor/acceptor ligands (**5 lectures**) Synthetic, characterization and reactivity strategies of Transition Metal hydrides bearing carbonyl, nitrosyl, carbene, phosphine, olefins, and so on as ancillary ligands. Inert bonds (C-H, C-C, O-H, N-H) and small molecules (H_2, CO, CO_2) activation mediated by organometallic complexes. (**12 lectures**)

Definition and concepts of catalysis: Important terminologies in catalysis Hydrogenation, dehydrogenation, dehydrogenative coupling reactions, hydrosilylation, hydroformylation, hydroamination, hydrocyanation, hydrozirconation, carbonylation, Wacker oxidation, Fischer Tropsch synthesis, water gas shift reaction, Olefin polymerization, olefin metathesis, Mechanistic pathway interpretation: Kinetic studies, Deuterium Kinetic Isotope effect, Rate limiting step, Resting state, Scrambling experiments. (**13 lectures**)

Approaches towards Sustainability using organometallic complexes: Pincer catalysts design, Utilization of CO_2 and other feed stocks; Sustainable Energy: H_2 economy, Methanol economy, Liquid organic hydrogen carrier. (**5 lectures)**

Frustrated Lewis pair Chemistry: Hydroboration, carboboration, small molecule activation, catalysis. (**4 lectures**)

Text Books

1. Crabtree, R. H. *The Organometallic Chemistry of the Transition Metals*, Wiley, 1998
2. Gupta, B. D.; Elias, A. J. Basic Organometallic Chemistry: Concepts, Syntheses and Applications Paperback, Universities Press; 2nd Edition (May 30, 2013).

Reference Books

1. Collman J. P.; Finke R. G.; Norton J. R. *Principles & Applications of Organo-transition Metal Chemistry*, University Science Books
2. Bhaduri S.; Mukesh, D. Homogeneous Catalysis: Mechanisms and Industrial Applications

Self Learning Material

S1. Maiti, D. *Organometallic Chemistry*, NPTEL Course Material, Indian Institute of Technology Bombay, 1. https://www.youtube.com/watch?v=ITfwjQemwMg&list=PLj_Alq7xw30l8iUgacWidP_83A0r ZqlBR&index=1

Title	**Catalysis for Energy**	Course No.	CYL6XX0
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	MSc Chemistry	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. Give an overview about defining a roadmap for the role of catalysis in energy production
2. Relate catalysis with fuel cells, hydrogen production and storage, methane storage and industrial catalysis

Learning Outcomes

The students are expected to have the ability to:

1. Understand the role and the possibilities of catalysis in the production of new energy carriers and in the utilization of different energy sources.
2. Go beyond conventional application identifying new developments that may lead to breakthroughs in the production of alternative energy.

Contents

Introduction: Global energy landscape & challenges, role of catalysis in energy conversion (**2 Lectures)**

Catalysis: Introduction to heterogeneous catalysis, including models of surface reactivity, surface equilibria, kinetics of surface reactions, electronic and geometrical effects in heterogeneous catalysis.(**Lectures 12**)

Catalytic reactions and application: Trends in reactivity, catalyst structure and composition, electro-catalysis and photo-catalysis, artificial photosynthesis, biomass conversion. (**Lectures 12**)

Selected applications and challenges in energy transformations: fuel cells, hydrogen production , water splitting, OER and HER reactions, CO2 oxidation, methane and hydrogen storage and industrial catalysis. (**Lecture 13**)

Textbooks

1. Hanefeld, U, Lefferts, L. (2018) Catalysis: An Integrated Textbook for Students, 1st Edition, Wiley

Reference Books:

1. Okada, Tatsuhiro, Kaneko, Masao, (2009), *Molecular Catalysts for Energy Conversion* Springer.
2. Guczi, László, Erdôhelyi, András, *Catalysis for Alternative Energy Generation*, Springer.
3. Pierluigi Barbaro, Claudio Bianchini, *Catalysis for Sustainable Energy Production,* Wiley

Self Learning Material

1. Xile Hu, Angamuthu, R and Bera, J. K., Catalysis for Energy Storage, GIAN Course material, Indian Institute of Technology Kanpur, https://www.youtube.com/watch?v=LvTfsIAI-VE

Title	**Characterization Techniques in Inorganic Chemistry**	Course No.	CYL7XX0
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	MSc/PhD (CY) Program	Type	Elective
Prerequisite			

Objectives

The instructor will:

Provide an introduction to various characterization methods that are applied to inorganic chemical systems.

Learning Outcomes

The students are expected to:

1. Explore different important characterization methods that are applied to inorganic chemical systems.
2. Analyse the chemical and physical characterization of molecular (solution and solid- state) inorganic chemical systems.

Contents

Fundamental aspects of characterization methods in inorganic chemistry.

Molecular Symmetry: Point group analysis of simple inorganic compounds. Infrared and Raman spectroscopy of simple inorganic molecules; predicting number of active modes of vibrations, analysis of representative spectra of metal complexes with various functional groups. (**6 Lectures**)

Electronic spectroscopy: Electronic absorption and emission spectroscopy (**4 Lectures**)

Multinuclear NMR: Use of 1H and 13C NMR to study fluxionality and dynamics in inorganic and organometallic chemical systems; NMR spectral analyses of B, Al, Si, F, and P containing compounds. Inorganic and Organometallic compounds of some other NMR active nuclei will also be discussed. (**11 Lectures**)

EPR: Elementary aspects of Electron paramagnetic resonance (EPR) spectroscopy of inorganic compounds- g-values, hyperfine and super hyperfine coupling constants; selected applications in inorganic chemistry. (**5 Lectures**)

Mass spectrometry: basic principles, ionization techniques, isotope abundance, molecular ion; illustrative examples from supramolecules, inorganic/coordination and organometallic compounds. (**5 Lectures**)

X-Ray and Photoelectron Spectroscopy: Electron microscopy and energy dispersive analysis of X-rays (**5 Lectures**)

Cyclic voltammetry: Introduction to cyclic voltammetry and discussion on cyclic voltammogram of some representative examples. (**3 Lectures**)

Text Books

1. R. S. Drago, Physical Methods for Chemists, 2nd Edition, Saunders, 1992.

Reference Books

1. M. Weller, Inorganic Chemistry, 6th Edition, Oxford University Press, 2014.
2. J. E. Huheey, Inorganic Chemistry, Principles of Structure and Reactivity, 4th Edition, Pearson, 2006.
3. F. A. Cotton, Advanced Inorganic Chemistry, 6th Edition, Wiley, 2007.

Self Learning Material

1. Alagarsamy P, *Characterization of Materials* NPTEL Course Material, Department of Chemistry, Indian Institute of Technology

Guwahati,https://nptel.ac.in/courses/115103030/

Title	**Inorganic Rings, Clusters and Polymers**	Number	CYL6XX0
Department	Chemistry	L-T-P [C]	3-0-0 [3]
Offered for	MSc Chemistry	Type	Elective
Prerequisite			

Objectives

The instructor will:

1. Give better insights into the interesting class of compounds such as rings, clusters, polymers and low valent complexes of the main-group elements.
2. Provide learning about the synthesis, structure and properties of inorganic heterocyclic and homocyclic rings.
3. Discuss structure, properties and reactivity of clusters and cages of main group elements.

Learning Outcomes

The students are expected to:

1. Appreciate the interesting class of compounds such as rings, clusters and polymers of main group metal compounds.
2. Gain deeper insights into the synthesis and reactivity of novel main group compounds.
3. Understand structure-property relationships and applications of inorganic, organometallic polymers and low valent compounds.

Contents

Introduction to p-block elements: Periodic trends and classification of p- block elements, Structure and bonding aspects. (**4 Lectures**)

Inorganic Rings: Synthesis, structure and reactivity of compounds containing heterocyclic rings of elements such as Phosphorus, Boron, Silicon, Tin and Aluminum. Structure and reactivity of compounds containing homocyclic rings of germanium, boron, silicon, aluminum. Synthesis and properties of inorganic macrocycles, crown and inverse-crown behavior. (**8 Lectures**)

Clusters: Clusters and Cages of Main Group Elements, Boron Hydrides and Carboranes. Homo and Heteropolyatomic Anions of the Post-Transition Elements. Soluble metallo-silicates and -phosphonates cages and clusters. (**8 Lectures**)

Inorganic polymers: Synthesis, structure-property relationships and applications of polysiloxanes, polyphosphazenes, polysilanes and organometallic polymers. (**10 Lectures**)

Applications: Main group element containing rings, clusters and polymers as single source precursors for nanomaterials. p-block elements in small molecule activation. (**9 Lectures**)

Textbook

1. Chivers, T. and Manners, I., (2009), Inorganic Rings and Polymers of the p-Block Elements, 1st Edition, Royal Society of Chemistry
2. P. Atkins, (2008), Inorganic Chemistry, Indian Edition, 4th Ed, Oxford University Press

References Books

1. H. R. Allcock, (2003), Chemistry and Applications of Polyphosphazenes, Wiley
2. V. Chandrasekhar, (2005), Inorganic and Organometallic Polymers, Springer-Verlag
3. M. Driess, (2004), Molecular Clusters of the Main Group Elements, Wiley-VCH

Self Learning Material

1. Balakrishna, M. S., *Chemistry of Main Group Elements*, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay, http://nptel.ac.in/courses/117101105/

Title	**Crystallography and X-ray Techniques for Chemists**	Number	CYL7XX0
Department	Chemistry	L-T-P [C]	3-0-0[3]
Offered for	MSc/PhD (CY) Program	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. Introduce the fundamental concepts of crystal symmetry and lattice structures and X-ray diffraction techniques are used for structure determination
2. Provide understanding of crystallographic data analysis and structure solution and discuss thermal analysis techniques for studying materials stability and phase changes

Learning Outcome

The students are expected to have the ability to:

1. Understand basic principles of X-ray diffraction, symmetry, lattices, and crystal structures.
2. Interpret diffraction data for crystal structure determination and understand methods used for crystal growth and materials characterization.

Contents

Introduction to Crystallography and X-ray Diffraction: Importance of crystallography in molecular and materials science. Basics of X-rays, detection and Monochromators. Bragg's law, Crystal systems, Bravais lattices. Concept of direct and reciprocal lattices. **(12 Lectures)**

Symmetry, Space Groups & X-ray Diffraction: Symmetry elements (rotational axes, screw axes, glide planes) point groups and space groups. Transitions into X-ray diffraction, including Bragg's law, experimental methods, diffractometry, and analysis of diffraction patterns. Determination of lattice parameters and crystal structure. **(15 Lectures)**

Application, Crystal Growth, Characterization & Analysis: Application of crystallography in materials science, pharmaceuticals and porous framework materials. Different techniques to grow crystals. Thermodynamics and kinetics of nucleation and crystal growth from solutions, melts, and vapors (CVD/PVD). Practical crystal structure solution. Powder X-ray analysis **(12 Lectures)**

Textbook

1. X-ray Structure Determination- A Practical Guide by G. H. Stout and H. L. Jensen, MacMillan, N.Y. 1968.
2. Contemporary Crystallography, M. J. Buerger, McGraw-Hill, N.Y. 1970.
3. Crystal Structure Analysis- A Primer J. P. Glusker and K. N. Trueblood, OUP, N.Y. 1985.
4. Structure Determination by X-ray Crystallography, M. Ladd and R. Palmer, Kluwer Academic/Plenum, N.Y. 2003.
5. Elements of X-ray Diffraction: BD Cullity

Self-Learning Material

1. Ray, R.K., X-ray Crystallography and Diffraction, NPTEL course material, Indian Institute of Engineering Science and Technology, Shibpur, Indian Institute of Technology, Madras, http://digimat.in/nptel/courses/video/112106227/L01.html

2. Choudhury, A. R., Chemical Crystallography, NPTEL Course Material, Department of Chemistry, Indian Institute of Science Education and Research, Mohali, https://nptel.ac.in/courses/104106093

Title	**Advance Fluorescence Microscopy Technique and Application**	Number	CYL7XX0
Department	Chemistry	L-T-P [C]	3-0-2 [4]
Offered for	MSc in Chemistry	Type	Elective
Prerequisite			

Objectives

1. Light microscopy has become one of the most useful tools in chemistry and life sciences. This course will cover basics and various methods of imaging fluoresces sample, image processing and lab experiments.

Learning Outcomes

1. Student will learn the fundamentals of light microscopy, theory of various fluorescence imaging methods.
2. Student will also learn how to do experiments with fluorescence microscope and image analysis.

Course Content

Introduction: History of light microscopy, What is light and color, Fourier space and Fourier transformation, Light as probe of matter, Properties of polarized light, Fluorescent dyes and proteins, Physics of optical components of a light microscope, Lenses and geometrical optics, Diffraction and interference in image formation, Diffraction and spatial resolution, Structured illumination, Point spread function (PSF), Numerical aperture, Resolution vs Super-resolution, Design of band-pass and dichroic filters for FRET, Bleed- through, Digital image: CCD vs EMCCD vs SCMOS vs PMT, Sensitivity vs Speed **(14 lectures)**

Microscopy: Fundamental theory and applications of Phase-contrast and Dark-field microscopy, Differential Interference contrast (DIC) microscopy, Epi-Fluorescence microscopy, Laser scanning confocal microscopy, Total Internal Reflection Fluorescence (TIRF), Two-photon microscopy, Light-sheet Microscopy, Lattice light- sheet microscopy, Super-resolution microscopy-Stochastic optical reconstruction microscopy, Video Microscopy for Single-molecule tracking and measuring dynamics, plasma membrane dynamics, DNA & RNA dynamics, single-molecule to Tissue imaging. **(19 lectures)**

Experiments: (1) DIY: Build your own microscope, (2) How to focus a sample, (3) Measurement of object size from image size, (4) Measurement of point spread-function (PSF), (5) Demonstration of Abbe Diffraction, (6) Setting up of Khelar Illumination, (7) Multi color image analysis and deconvolution by Image]. **(6 lectures)**

Text Books:

1. Murphy, D. Fundamentals of Light Microscopy and Electronic Imaging (2001), Wiley & Sons, Inc.
2. Cornea, A and Conn, P. M. Fluorescence Microscopy (2014), Elsevier

Reference Books:

1. Lakowicz, J.R., (2006), Principle of Fluorescence Spectroscopy, 3rd edition, Springer, New Yor

Online Course Material:

1. iBiology Microscopy Series, Ron Vale, UCSF (https://www.ibiology.org/online-biology-courses/microscopy-series/)

Title	**Chemical Binding**	Number	CYL6XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry/BS in Engineering Chemistry	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. Provide high level understanding of a chemical bond using quantum mechanics.
2. Describe knowledge of modern techniques in quantum chemistry

Learning Outcomes

The students are expected to have the ability to:

1. Use modern quantum chemistry techniques to solve fundamental problems related to basic chemistry
2. Understand a chemical bond

Contents

Electronic structure: Variation theory, perturbation theory, electron spin, He atom, Pauli exclusion principle, Slater determinant (10 lectures)

Electronic structure of diatomic molecules: Born Oppenheimer approximation, Atomic units, H_2^+ ion, H_2 molecule, VB theory, MO theory, Semi-empirical methods (10 lectures)

Wave function based theory: Hartree-Fock theory, Electron correlation, Self-consistent field wave function, Moller-Plesset theory, Coupled cluster formulation (15 lectures)

Density functional theory (DFT): Principles of density functional theory, commercial functionals and applications (4 lectures)

Textbook

1. Levine, I. N., *Quantum Chemistry*, Pearson education (2000).
2. Lowe, J. P., *Elementary Quantum Chemistry*, Academic Press.

Reference Books

1. Szabo, A. and Ostlund, N. S., *Modern Quantum Chemistry*, Dover (1996).

Title	**Molecular Dynamics Simulations**	Number	CYL7XX0
Department	Chemistry	L-T-P [C]	3–0–0 [3]
Offered for	M.Sc. Chemistry	Type	Elective
Prerequisite			

Objectives

The Instructor will:

1. Provide high level understanding of the connection between dynamics of molecules and their macroscopic properties
2. Describe knowledge of modern techniques in molecular simulations using high speed computers

Learning Outcomes

The students are expected to:

1. Use modern simulation techniques to solve fundamental problems related to interdisciplinary areas of Physics, Chemistry and Biology
2. Find explanations for phenomena related to soft condensed matter

Contents

Basics of Molecular Dynamics: Theoretical foundation of classical statistical mechanics, Boltzmann's relation, harmonic oscillator and harmonic baths, introduction to molecular dynamics, Algorithms for equations of motions (9 lectures),

Various ensembles: Purpose of ensembles in simulations, canonical ensembles, isobaric ensembles, grand canonical ensembles, applications of ensembles, thermostat, Barostat (10 lectures)

Advance Techniques: Free energy perturbation theory, Thermodynamic Integration, Jarzynski's equality, Umbrella Sampling (10 lectures)

Applications: How to set up Membrane system, protein and small biological molecules (10 lectures)

Textbooks

1. D. Frenkel and B. Smith, (2002), *Understanding Molecular Dynamics Simulations*,1st Edition, Academic Press
2. Mark Tuckermann, (2010), *Statistical Mechanics and Molecular Simulations*, 1st Edition, Oxford University

Self Learning Material

1. M. P. Allen and T. J. Tildesley., (1993), *Computer Simulations of liquids*, 1st Edition, Oxford Science Publications

Preparatory Course Material

1. Banerjee, T. *Molecular simulations*, NPTEL Course Material, Department of Chemical Engineering, Indian Institute of Technology Guwahati, http://nptel.ac.in/courses/103103036/
2. Tembe, B.L., *Computational Chemistry and Classical Molecular Dynamics*, NPTEL Course Material, Department of Chemistry, Indian Institute of Technology Bombay, https://onlinecourses.nptel.ac.in/noc18_cy13

Title	**Modern NMR Methods for Chemists**	Course No.	CYL6XXX
Department	Chemistry	Structure (L-T-P [C])	3–0–0 [3]
Offered for	M.Sc. in Chemistry	Type	Elective
Prerequisite			

Objectives

1. To introduce the fundamentals of Nuclear Magnetic Resonance with an emphasis on the modern methods
2. Introducing one and two dimensional NMR methods applicable in research in Chemistry and allied areas

Learning Outcomes

The students are expected to have the ability to:

1. Understand the basic principle of Nuclear Magnetic Resonance.
2. Appreciate different NMR methods in solution state with their advantages and limitations.
3. Acquire the knowledge of recording and processing NMR spectral data
4. Apply NMR methods for analyzing molecular systems relevant in Chemistry, materials and biology.

Course Contents

Introduction to Magnetic Resonance: Basic concepts of spin resonance, Zeeman interaction and resonance absorption, Bloch equation and relaxation, introduction to first order NMR spectrum, analysis of chemical shifts and coupling patterns, factors affecting NMR spectra, steady state and transient NOE, distance measurement and sensitivity enhancement, spectral editing. (13 lectures)

Introduction to one dimensional NMR experiments: Radiofrequency pulse and Fourier transformation, selective excitation and inversion, solvent suppression techniques, measurement of relaxation and diffusion, introduction to standard 1D NMR methods such as inversion recovery, CPMG, NOE , ROE, INEPT, DEPT and STD NMR. (13 lectures)

Introduction to two dimensional NMR experiments: separation of interaction, 2D J-resolved spectroscopy, correlation spectroscopy –COSY and TOCSY, 2D NOE experiments and exchange spectroscopy, diffusion ordered spectroscopy, Magic Angle Spinning (MAS) experiments with cross-polarization in solid state NMR, Analysis of 1D and 2D NMR spectra of some selected molecules. (13 lectures)

Textbook

1. Gunther, H., (1995), NMR Spectroascopy, 2nd Edition, Wiley
2. Derome, A. E., (1987), Modern NMR Techniques for Chemistry research, 1st Edition, Pergamon Press

Reference Books

1. Farrar, T.C. and Becker, E. D., (1971), Pulse and Fourier Transform NMR, 1st Edition, Academic Press
2. Claridge, T. D. W, (2016), High resolution NMR Techniques in Organic Chemistry, 3rd Edition, Elsevier.
3. Sanders, J.K.M, Hunter, B.K., (1993), Modern NMR spectroscopy-A guide for chemists, 2nd Edition, Oxford English monographs.

Online course material:

1. Kumar, A and Hosur, R. V., (2019) NMR Spectroscopy for Chemists and Biologists, IIT Bombay. https://nptel.ac.in/courses/104/101/104101117/

2. Suryaprakash, N., (2020) One and two dimensional NMR Spectroscopy for Chemists, IISc. Bangalore. https://nptel.ac.in/courses/104/108/104108124/.

Course Title	**Principles of Biological Soft Matter**	Course No.	CY6XXX
Department	Chemistry	L-T-P[C]	3-0-0 [3]
Offered for	BS in Engineering Chemistry, M.Sc. in Chemistry	Type	Elective
Pre-requisite			

Objectives

The instructor will:

To understand basic principles and theory of biological soft matter

To understand structure-property relationship of biological soft matter

Learning Outcomes

The students are expected to:

1. integrate concepts from biology, chemistry, physics, and mathematics to understand biophysics
2. connect structure, simulation, and functions of protein, membrane

Course contents

Fundamentals of Soft Matter: time, length and energy scales of soft matter; intermolecular forces; macromolecules; colloids, liquid crystals, amphiphilc self-assembly (10 lectures)

Soft matter in Nature: Nucleic acids; structure, folding, interactions of proteins; polysaccharide, Different lipids and membrane structure, functions, Membrane Enzymes, Receptor, Transporter, channel, water, Active matter in living system (15 lectures)

Statistical Mechanics and soft matter: Entropy, 2nd law of thermodynamics, energy, temperature, free energy, distribution functions; random walk, fluctuations, simulations, introduction to the theory of active matter (14 lectures)

Textbooks

1. Jones, R. A. L., Soft Condensed Matter, Oxford University Press 2002.
2. Jacob N. Israelachvili, Intermolecular and Surface Forces, Academic Press, 2011.
3. Anders Liljas, Lars Liljas, Miriam-Rose Ash, Goeran Lindblom, Poul Nissen, Morten Kjeldgaard, Textbook of Structural Biology, World Scientific, 2020.

Reference Books

1. Mary Luckey, Membrane Structural Biology, Cambridge University Press, 2013.
2. Alexander Yu. Grosberg and Alexei R. Khokhlov, Statistical Physics of Macromolecules, AIP Press, 1994
3. M. P. Allen and D. J. Tildesley, Computer Simulation of Liquids, Clarendon Press, 1991.

Online Course Material

1. NPTEL course by Profs. Sunil Kumar, P. B., and Menon, G. I.,; "Physics of Soft Condensed Matter"; https://nptel.ac.in/syllabus/115106063/
2. Membrane BioPhysics and Membrane Proteins, https://www.youtube.com/watch?v=Fict6w40dJ0